Improving forage production in Norway: Participatory evaluation of climate-adapted seed mixtures

Jørgensen M.[1], Borchsenius R.[2], Elverland E.[1], Gillund F.[1], Muradagha K.[1], de Sousa K.[3] and Sturite I.[1]

[1] Department of Grassland and Livestock, NIBIO, Norwegian Institute of Bioeconomy Research, Ås, Norway; [2] NLR, Norwegian Advisory Service, Ås, Norway; [3] University of Inland Norway, Lillehammer, Norway

Abstract

A stable supply of high-quality forage is essential for the economic sustainability of milk and meat production, the most important agricultural sectors in Norway. Successful forage production relies on selecting species, cultivars, and seed mixtures suited to the diverse agroecological conditions, which vary widely in latitude and altitude. Climate change modifies production boundaries unpredictably. Without on-farm performance data, there is a risk of recommending new varieties or seed mixtures that underperform in farmers' fields. To address this, we propose the *tricot* method: a participatory, large-scale, and cost-effective testing approach where farmers test new seed mixtures directly in their fields. This approach involves many farmers in conducting small, simple experiments. Each trial consists of an incomplete block with three options out of the total ten options in the trial. Farmers evaluate these options based on predetermined criteria using digital tools. These mini trials, combined with data on management, climate, and soil, are collected into the ClimMob software, which performs statistical analyses and provides feedback. When all data are aggregated, this approach can provide robust statistical results on optimal seed mixtures for various local conditions. This method also strengthens collaboration between researchers, advisers, and farmers, contributing to more sustainable and resilient agricultural practices.

Keywords: climate adaptation, collaboration, digital, field trials, local conditions, participatory

Introduction

Traditionally, Randomised Complete Block Design (RCBD) field trials have been used to assess new forage species, cultivars, and mixtures across diverse agroecological zones. However, these trials are labour-intensive and costly, limiting the number of sites and often excluding farmers from the evaluation process, which can reduce practical relevance and adoption. Participatory research addresses this by involving farmers in setting goals, conducting trials, and evaluating outcomes, ensuring that research is tailored to local conditions, and allows for innovation, learning, and ownership (van de Gevel et al. 2020). On-farm experimentation also helps integrate scientific and farmer knowledge (Lacoste et al. 2022). The tricot (triadic comparison of technologies) approach is a promising participatory method. Each farmer receives an incomplete block of three options out of a larger pool, ranks them using the best-worst scaling approach, and submits results to the free online software platform ClimMob (https://climmob.net/). This incomplete block design enables large-scale, cost-effective data collection, and rapid feedback across varied environments. Although tricot has been applied in several studies in the Global South, it is new to Norway. The INNOFORAGE project (2025–2028) aims to establish a cost-effective, farmer-led testing framework to identify optimal forage mixtures for local conditions and promote adoption. We will evaluate if tricot can deliver robust results across Norway's diverse climates and integrate it with RCBD trials at research station and selected farms. While tricot enables broad participation, metric traits like feed quality and biomass yield still require controlled

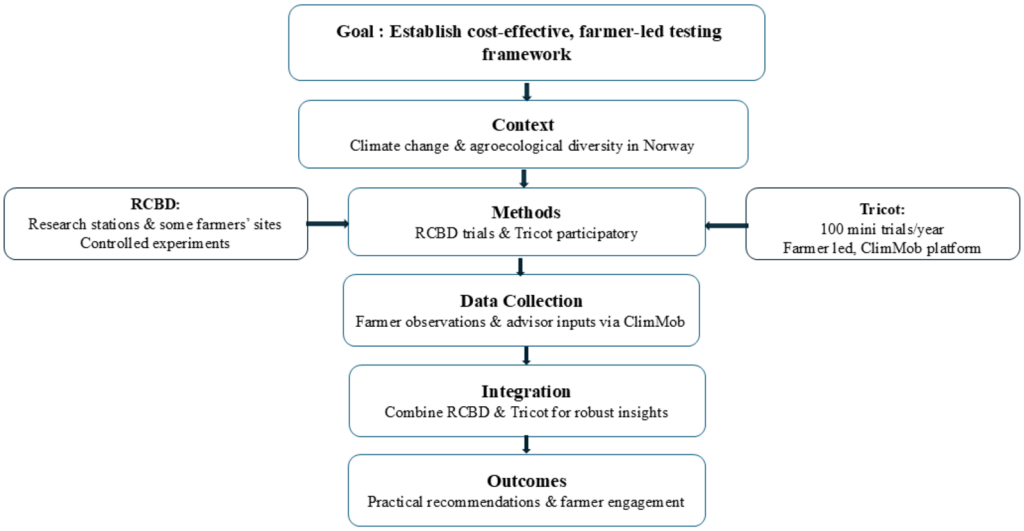


Figure 1. Flow chart of the different activities and work within the project.

experiments. By combining both approaches, we aim to provide more targeted recommendations and assess how participatory research supports farmer engagement and knowledge sharing.

Materials and methods

A flow chart of the work in the project is presented in Figure 1.

Twelve forage seed mixtures were developed in collaboration with seed companies, advisers and researchers (Table 1). Using the ClimMob platform, three seed mixtures were randomly assigned to each mini-trial via a balanced incomplete block design. In summer 2025, 100 mini-trials were established on farms across Norway, each with three 3×7 m^2 plots managed as part of the field by farmers. Management practices, soil characteristics, and agro-climate indices are recorded for each trial and included as covariates in the confirmatory Plackett-Luce model to reduce confounding. The primary outcomes are overall performance rank, winter survival, and yield, with key comparisons across mixture × region × year. Secondary outcomes (e.g., weed invasion, plant height, legume cover) are analysed exploratorily. Farmers have provided informed consent and agreed to take part in focus group interviews and surveys regarding their experience with the experimentation. In the establishment year, they assess the three options in autumn based on: cover of sown species and of legumes (which is best/which is worst), weed invasion (least/most), plant height (tallest/shortest). In spring 2026, they will also report on winter conditions, particularly ice cover on fields. At harvest, they will assess yield (most/least), proportion of legumes (most/least), and proportion of weeds (least/most). In addition, they will be asked on overall performance compared to surrounding field (best/worst). They also will provide information on fertilisers (type and quantity). Trials will run for at least two full production years. Another 100 participants will be recruited in 2026, and seed mixtures may be adjusted based on feed-back from farmers and advisors participating in round one.

ClimMob manages all participant and data processes, supports rapid feedback, and includes R packages for statistical analyses and agro-climatic indices (Quiros et al. 2024). Farmer engagement and

Table 1. Overview of seed mixtures

Mixture	*Phleum pratense*	*Lolium pratense*	*Poa pratensis*	*Dactylis glomerata*	*Bromus inermis*	*L. arundinaceum*	*Festuca rubra*	*Agrostis capillaris*	*L. perenne*	*Trifolium pratense*	*T. repens*	*Medicago sativa*	Region
1	70	20	0	0	0	0	0	0	0	10	0	0	Common
2	50	20	20	0	0	0	0	0	0	0	10	0	Common
3	25	10	15	0	0	25	5	0	10	5	5	0	Common
4	10	15	10	0	15	20	5	2	8	10	5	0	Common
5	0	30	0	40	0	0	0	0	10	10	10	0	Common
6	10	0	0	20	0	50	0	0	10	10	0	0	Common
7	10	0	0	0	0	70	0	0	10	10	0	0	Common
8	40	50	0	0	0	0	0	0	0	10	0	0	South/Central
9	40	50	0	0	0	0	0	0	0	10	0	0	North/Mountain
10	25	0	10	0	0	45	0	0	0	10	0	10	Common
11	0	0	0	0	0	0	0	0	90	0	10	0	South/Central
12	45	15	15	0	0	0	5	5	0	10	5	0	North/Mountain

learning will be studied using mixed methods, including interviews, surveys and stakeholder meetings throughout the project period.

Results and discussion

The mini-trials are currently being sown by advisers and farmers. Of the 100 mini-trials planned for 2025, 30 with seed mixtures targeted for northern and mountainous regions have been distributed to farmers located c. 61–70°N. The remaining 70 trials, designed for central and southern sites, have been distributed to farmers between about 59 and 67°N. After all participating farmers are registered, they will receive an initial set of questions regarding the establishment of their seed-mixture plots. We expect the first results to be available in early autumn 2025. Farmers have expressed great interest in taking part in the experimentation. Interest in participating has exceeded expectations, with more than the 100 planned farmers expressing they would like to join. These additional participants will be included in the 2026 round of trials. The advisers have also shown enthusiasm, particularly in using the mini-trials in field days—events where farmers meet, share experiences and discuss results. This project will both support the adaptation of seed mixtures to local agroecological conditions, and improve adoption, ownership, and active engagement among all actors—especially the farmers.

Acknowledgements

The authors would like to thank the Agricultural Research Funds (FFL/JA), Grant no 352917 for financing the INNOFORAGE project.

References

Ergon Å, Seddaiu G, Korhonen P, Virkajärvi P, Bellocchi G, Jørgensen M. and Volaire F. (2018). How can forage production in Nordic and Mediterranean Europe adapt to the challenges and opportunities arising from climate change? *European Journal of Agronomy* 92, 97–106. 10.1016/j.eja.2017.09.016.

Lacoste M., Cook S., McNee M. Gale D., Ingram J., Bellon-Maurel V., …Hall A. (2022). On-Farm Experimentation to transform global agriculture. *Nature Food* 3, 11–18. 10.1038/s43016-021-00424-4.

Quiros C., de Sousa K., Steinke J., Madriz B., Laporte M. A., Arnaud E., . . . van Etten J. (2024). ClimMob: Software to support experimental citizen science in agriculture. *Computers and Electronics in* Agriculture 217 108539. 10.1016/j.compag.2023.108539.

van de Gevel J., van Etten J. and Deterding S. (2020). Citizen science breathes new life into participatory agricultural research. A review. *Agronomy for Sustainable Development* 40, 35. 10.1007/s13593-020-00636-1.